

Python-Taxis Data Analysis & Visualization

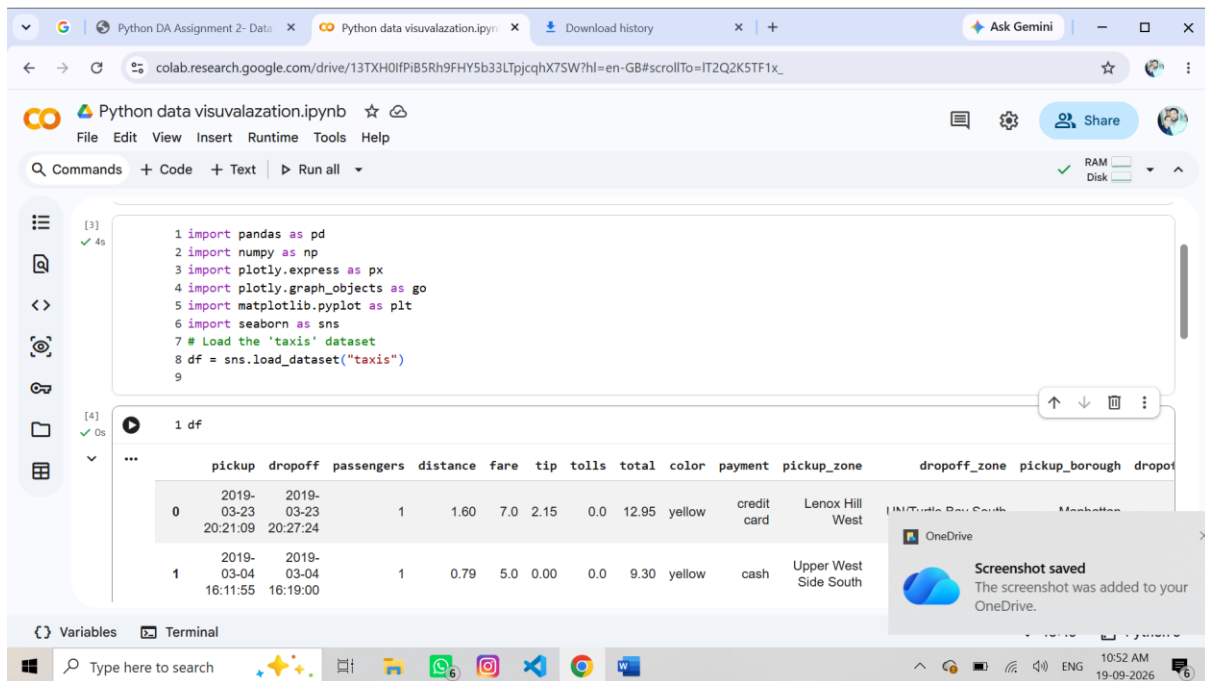
Objectives:

The objective of this project is to analyze the Seaborn Taxis dataset using Python, Pandas, Matplotlib, and Seaborn. The project focuses on handling missing values and creating different visualizations to understand taxi fares, trip distances, payment methods, tips, and pickup locations.

Tasks

- 1) Loading the Taxis Dataset using the following code:

```
import seaborn as sns
# Load the 'taxi' dataset
df = sns.load_dataset("taxi")
```



The screenshot shows a Google Colab notebook titled "Python data visuvalazation.ipynb". The code cell [3] contains the following Python code:

```
1 import pandas as pd
2 import numpy as np
3 import plotly.express as px
4 import plotly.graph_objects as go
5 import matplotlib.pyplot as plt
6 import seaborn as sns
7 # Load the 'taxi' dataset
8 df = sns.load_dataset("taxi")
9
```

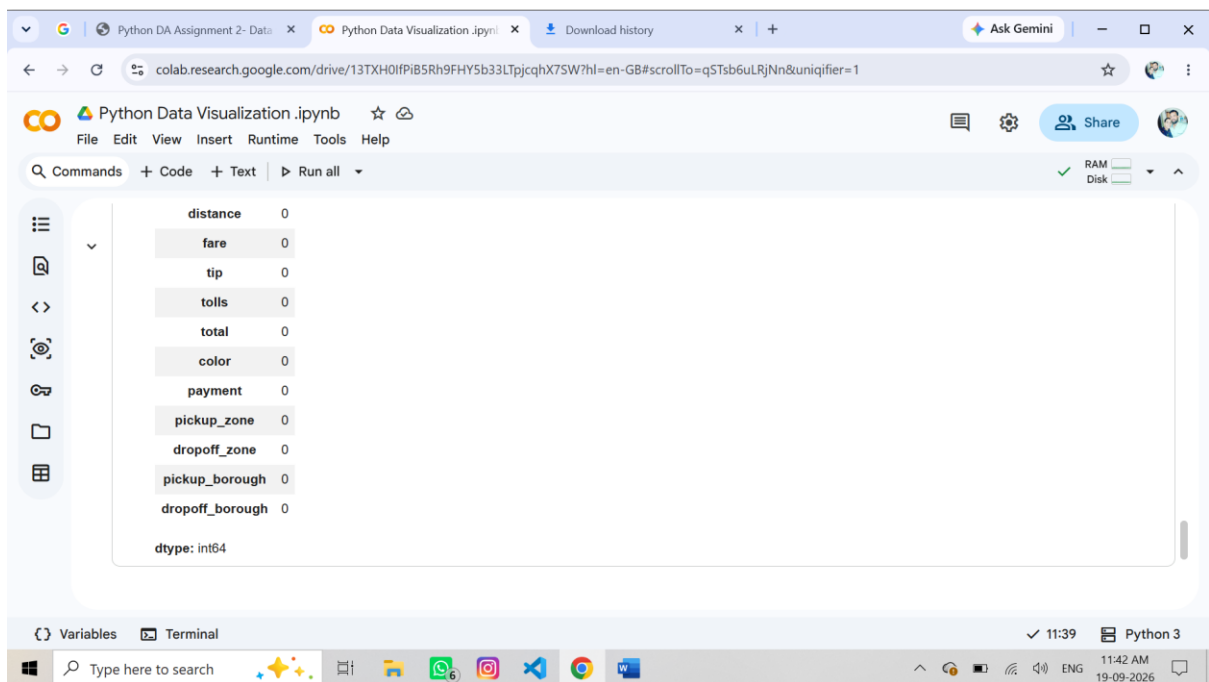
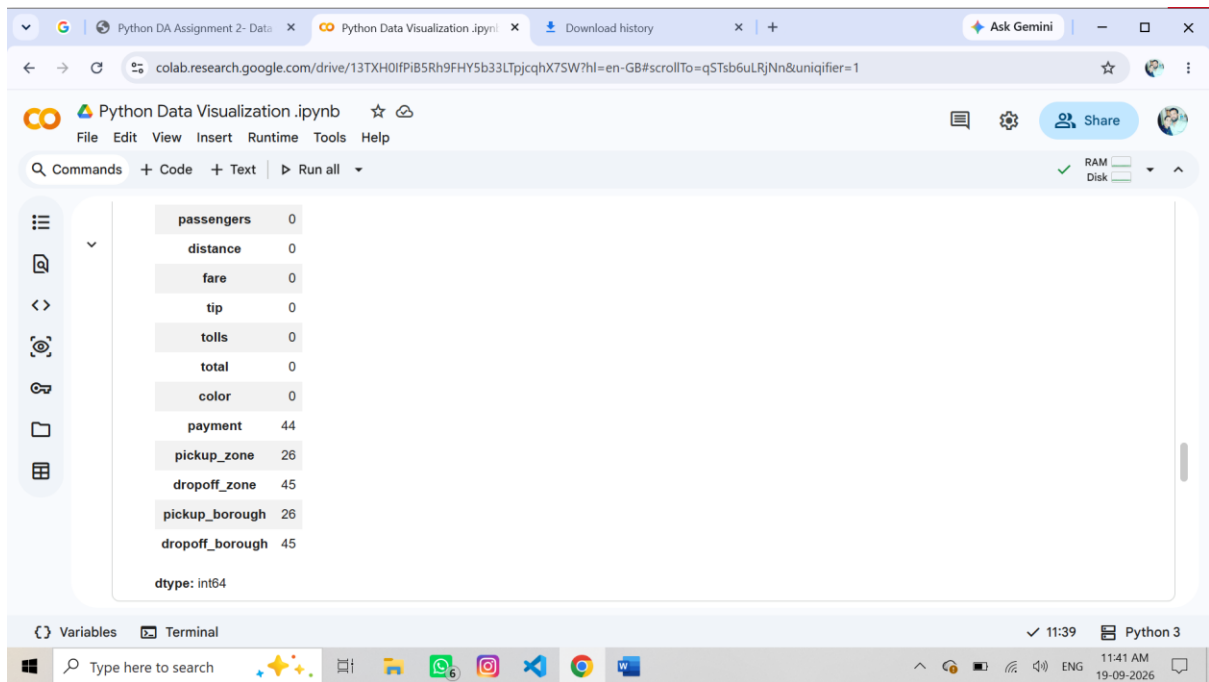
The output cell [4] displays the first two rows of the dataset as a table:

	pickup	dropoff	passengers	distance	fare	tip	tolls	total	color	payment	pickup_zone	dropoff_zone	pickup_borough	dropoff_borough
0	2019-03-23 20:21:09	2019-03-23 20:27:24	1	1.60	7.0	2.15	0.0	12.95	yellow	credit card	Lenox Hill West	Midtown East	Manhattan	Manhattan
1	2019-03-04 16:11:55	2019-03-04 16:19:00	1	0.79	5.0	0.00	0.0	9.30	yellow	cash	Upper West Side South	Midtown East	Manhattan	Manhattan

A OneDrive notification is visible in the bottom right corner: "Screenshot saved. The screenshot was added to your OneDrive."

- 2) Handling Missing Values

- Check for missing values in the dataset and identify columns with missing data.
- Impute missing values using appropriate strategies based on the column type (e.g., using the mean, median, or mode for numerical columns and the mode for categorical columns).
- For columns that are critical and cannot be reasonably imputed, remove rows with missing values to maintain data integrity.



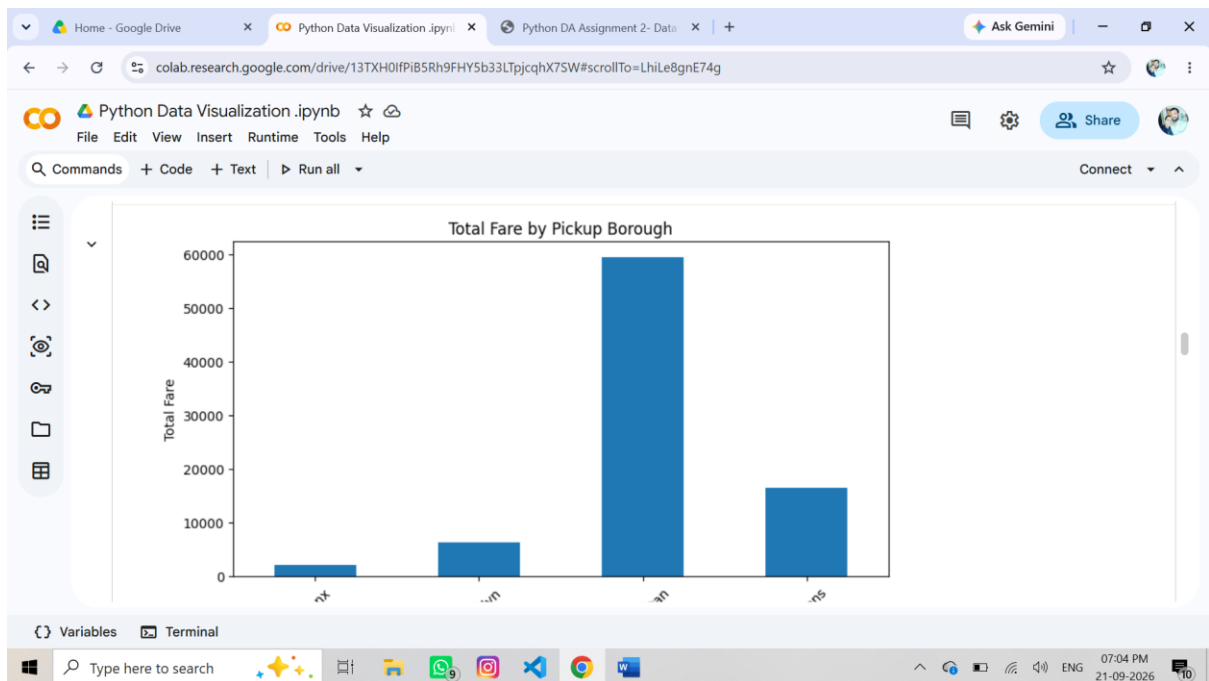
3) Visualizations using Matplotlib/Pandas Plot:

★ **Line Chart Plot** a line chart to visualize the fare over time, using the pickup timestamp as the x-axis and fare as the y-axis. Ensure the pickup column is converted to a datetime format before plotting.



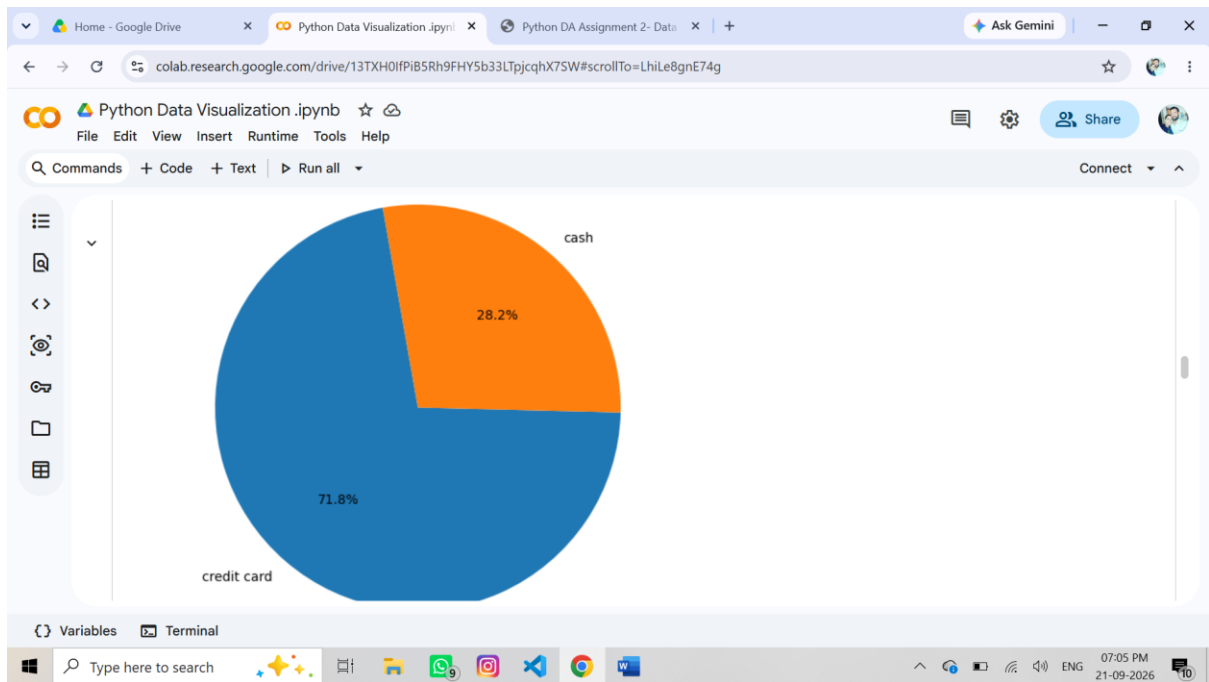
★ Bar Chart

Create a bar chart to show the total fare for each pickup_borough. Group the data by pickup_borough and sum the fare for each group.



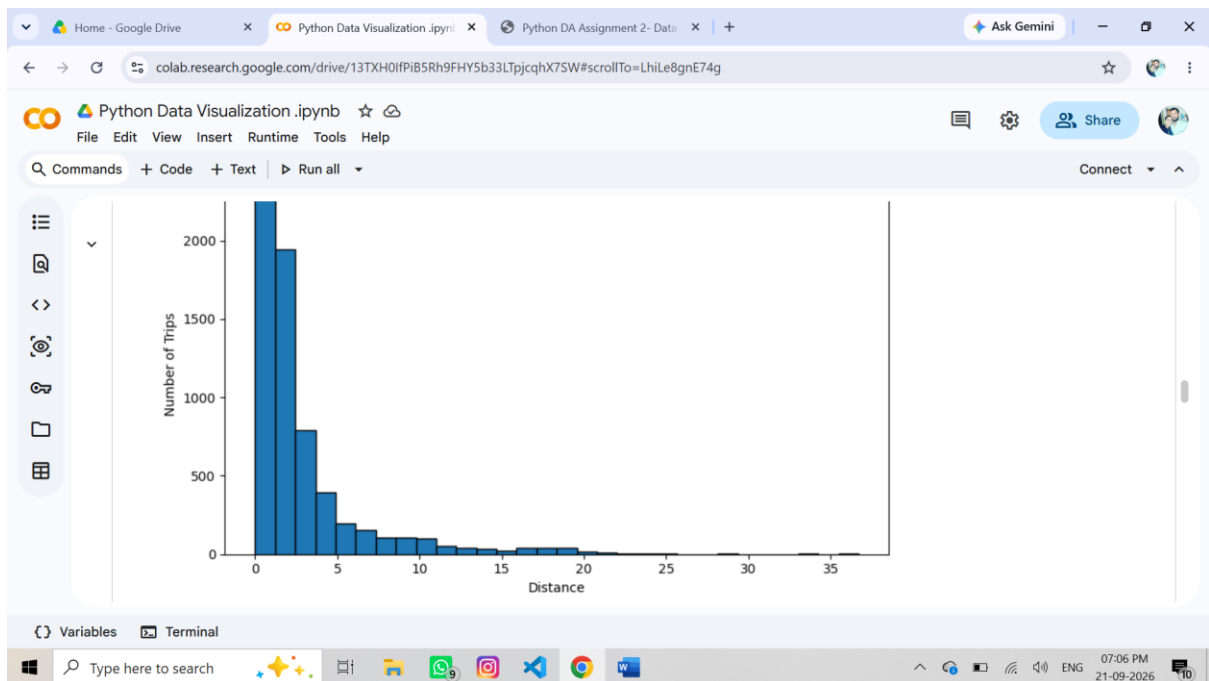
★ Pie Chart

Plot a pie chart showing the distribution of trips based on the payment method (credit card, cash, etc.). Each slice should represent the count of trips for a specific payment method.



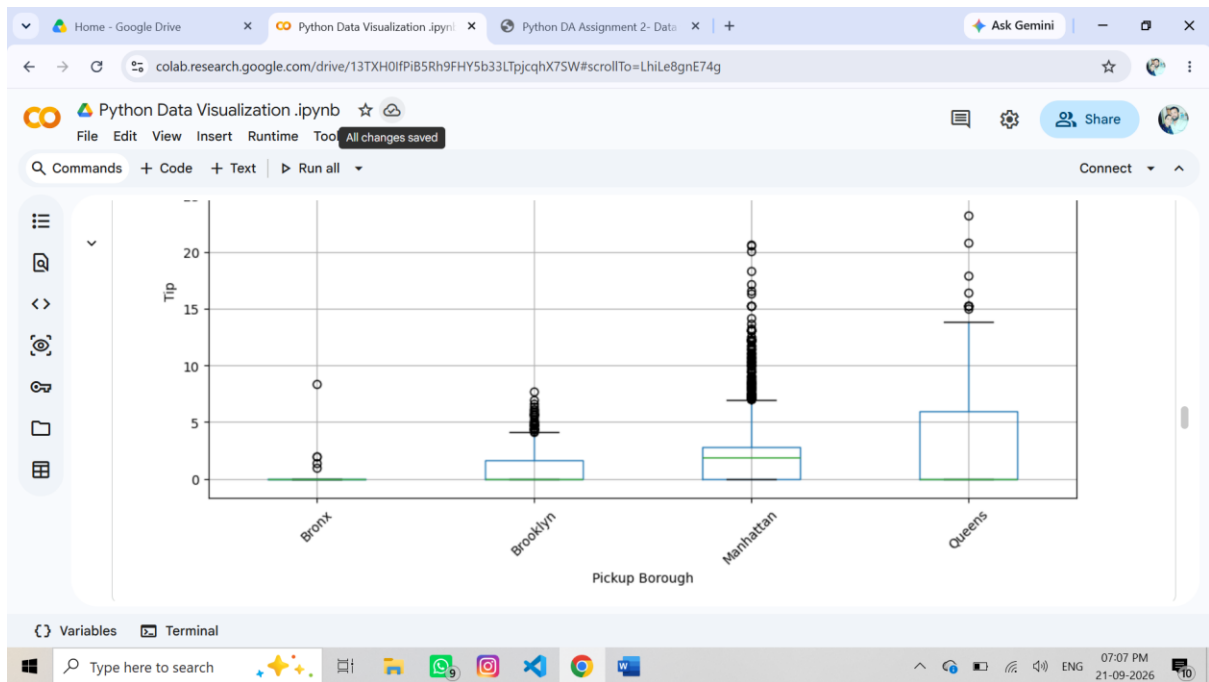
★ Histogram

Create a histogram to visualize the distribution of distance. Customize the number of bins for better granularity and ensure the plot is easy to interpret.



★ Box Plot

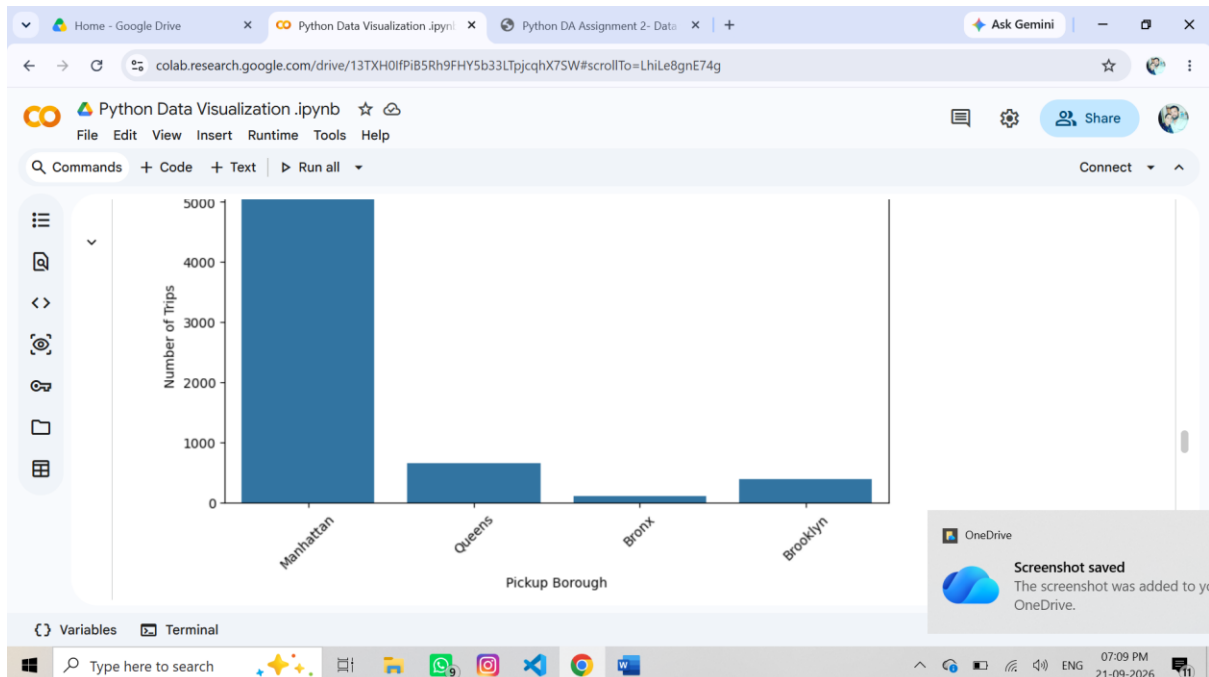
Plot a box plot to visualize the distribution of tip amounts for each pickup_borough. Use pickup_borough as the categorical axis and tip as the numeric axis.



Visualizations using Seaborn:

★ Count Plot

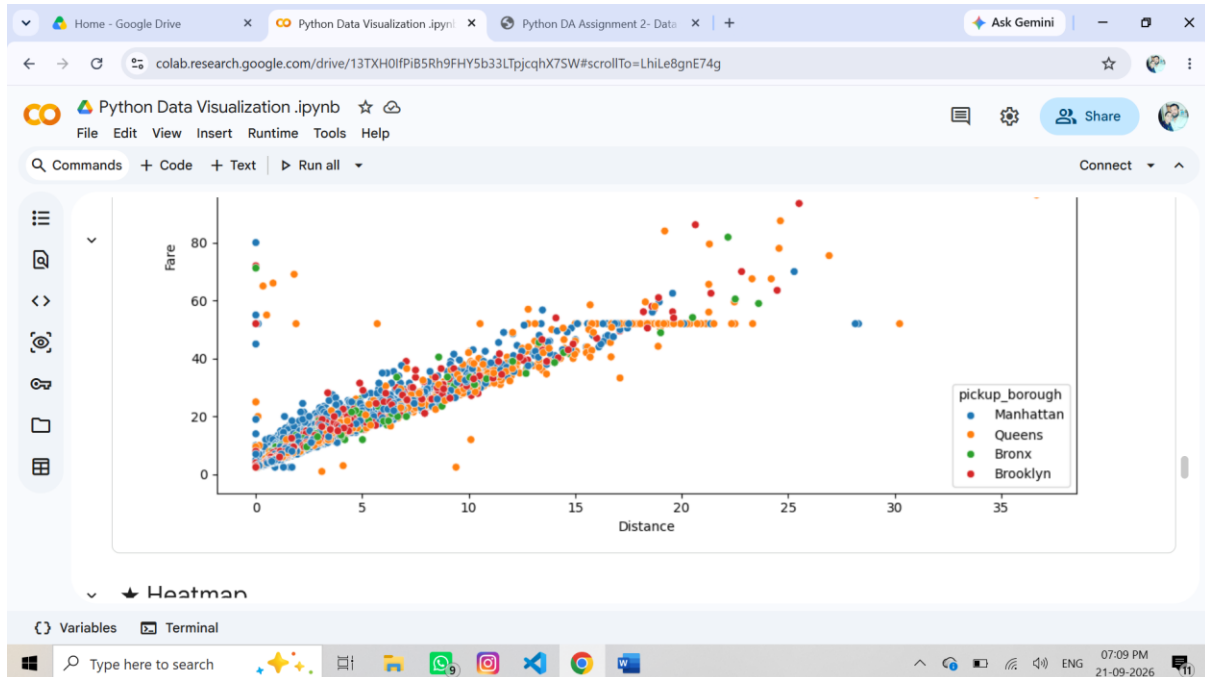
Create a count plot to visualize the number of trips in each pickup_borough. The x-axis should represent the boroughs, and the y-axis should show the count of trips



★ Scatter Plot

Plot a scatter plot to show the relationship between distance and fare. Use distance on the x-axis and fare on the y-axis to visualize any correlation. Color the points based on the pickup_borough to

differentiate the trips by their respective boroughs.



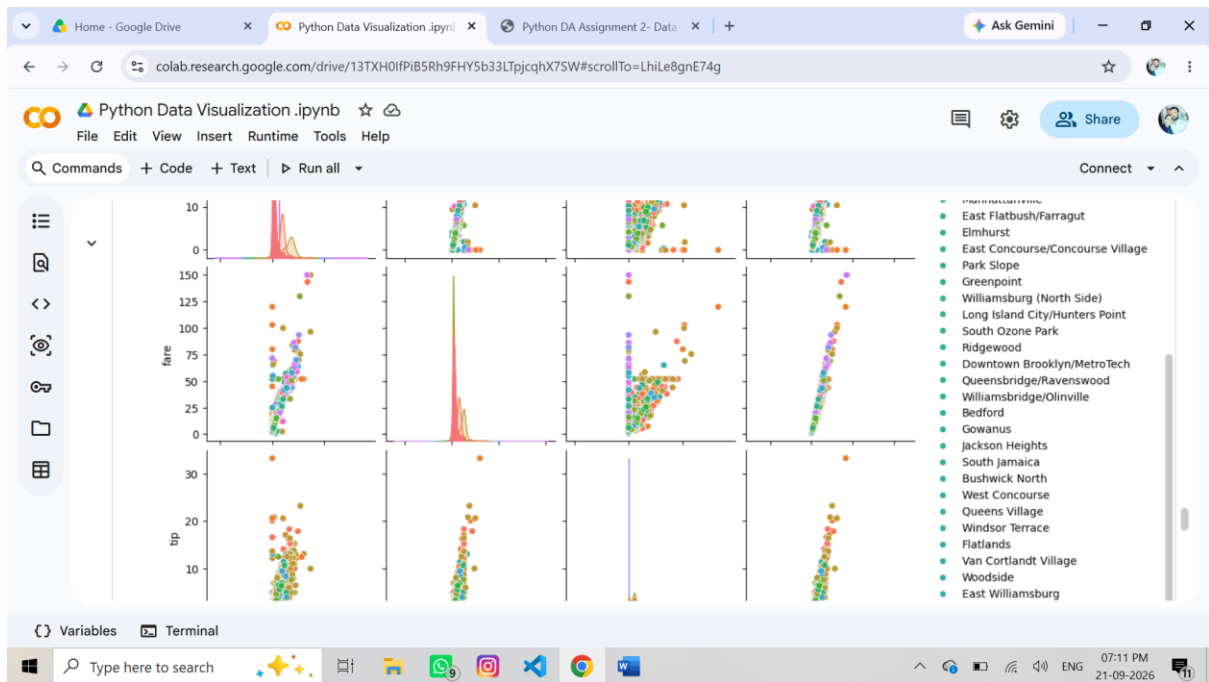
★ Heatmap

Plot a heatmap to visualize the correlation between numerical variables such as distance, fare, tip, tolls, and total. Use a correlation matrix to highlight the relationships.



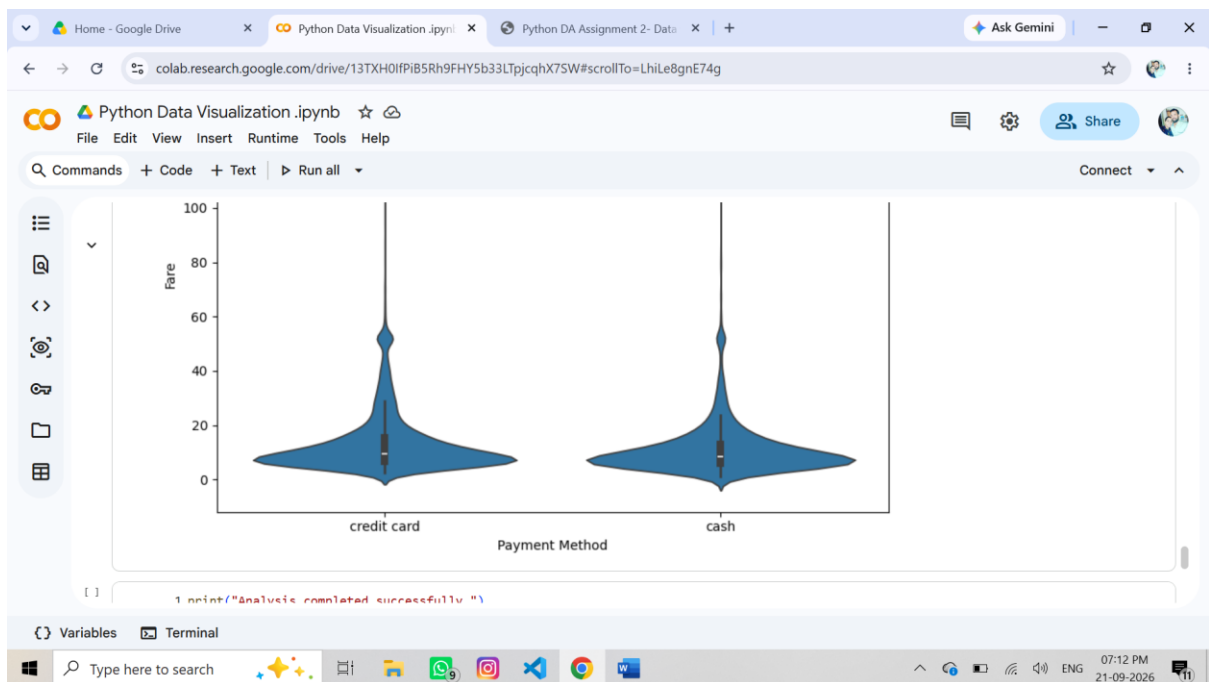
★ Pair Plot

Create a pair plot to visualize the pairwise relationships between distance, fare, tip, and total. Color the data points according to the pickup_zone column. to compare how different zones affect these variables.



★ Violin Plot

Plot a violin plot to show the distribution of fare for each payment method. Use the payment method as the categorical axis and fare as the numeric axis to visualize its distribution.



Analysis:

The visualizations help interpret taxi trip patterns by analyzing fare, distance, tips, payment methods, pickup boroughs, and pickup zones. The charts show how fares vary over time, how trip distance is distributed, which payment methods are commonly used, and how fares relate to distance. The box plot and violin plot help understand the spread of tips and fares, while the heatmap and scatter plot show relationships between numerical variables. Overall, the analysis provides a clear understanding of taxi trip behavior and fare patterns.

Conclusion:

The Taxis dataset was successfully cleaned and analyzed using Python. Different visualizations helped understand taxi trip patterns, fare distribution, distance, tips, payment methods, and pickup locations. This project improved practical knowledge of data cleaning, Pandas, Matplotlib, Seaborn, and data visualization, which are important skills for a Data Analyst.